Nguyen Hung Thien Loc

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Professional Summary

Data Engineer with two internships building and operating ETL/ELT pipelines and data warehouses. Built a PostgreSQL data warehouse consolidating SAP B1 and POS data for a 7-brand retail group, with data reconciliation, data quality checks and Grafana monitoring/alerting; previously built REST API ingestion pipelines and Python APIs for financial market data. Strong in Python and SQL; works independently and with business teams to deliver reliable data for BI dashboards and management reporting.

Experience

Thinh The Vinh Hoa **September 2026 – Present**
Data Engineer Intern *Ho Chi Minh City*

- Designed and built a **PostgreSQL data warehouse** (raw → staging → warehouse → datamarts, with **fact/dimension tables**) and **30+ automated ELT pipelines** consolidating **SAP B1** and **iPOS** data from **7+ source databases** for a 7-brand retail group, replacing manual extraction with **95% on-schedule runs**.
- Performed **data reconciliation** between the warehouse and source systems (SAP HANA, POS sales tables with ∼**200K rows/day** per brand), detecting **200 mismatches** and building automated heal workflows to correct them.
- Deployed **5+ Grafana dashboards** and **9+ real-time alerts** for infrastructure and pipeline health, reducing incident detection time to **under 1 minute**.
- Gathered requirements from business teams and delivered **2+ self-service Metabase dashboards** with automated daily reports, cutting manual reporting time by **85% (2 hours/day)**.
- Authored **technical documentation** in Markdown and a structured comparison of two warehouse architectures (SQL Server staging vs. direct CDC to PostgreSQL) to support the team's design decision.

Gradient Technologies **March 2026 – June 2026**
Data Engineer Intern *Ho Chi Minh City*

- Built asynchronous ingestion pipelines in **Windmill** pulling market data from **10 financial REST APIs**, with rate-limit handling, retries and **deduplication**, achieving **99% scheduled run success**.
- Developed and deployed **20+ Python APIs** serving downstream analytics teams with end-to-end latency **under 500 ms**.
- Refactored the monolithic codebase with strict Python typing and modular design, reducing debugging time by **30%**.

Projects

Real-Time Fraud Detection Platform | *Go, Kafka, Redis, gRPC, Kubernetes* | ***GitHub*** **February 2026**

- Built a pre-authorization fraud-blocking platform: **Go** service, **XGBoost** inference over **gRPC/Protobuf**, **Redis** balance caching, async **Kafka** write-behind to **PostgreSQL**, autoscaling on **Kubernetes**.
- Sustained **300 concurrent TPS** (>18,000 tx/min) with **100% success rate** and **836 ms** end-to-end latency, within the <2 s banking compliance threshold; full **Grafana** observability.

Technical Skills

Languages: Python, SQL, Go, C++
Data Pipeline & Integration: ETL/ELT, REST API, n8n, Windmill, Kafka, Apache Airflow
Data Warehousing & Modeling: PostgreSQL, Microsoft SQL Server (MSSQL), dbt
Monitoring & BI: Grafana, Prometheus, Metabase, Power BI
DevOps & Tools: Git, GitHub, Docker, Kubernetes, Linux (Ubuntu), Redis
Soft Skills: analytical thinking, problem-solving, independent work, cross-team collaboration
English: TOEIC 675

Awards

- **Bronze Medal (Top 88/235):** Vietnamese Olympiad in Informatics **(2024)**
- **Bronze Medal:** 30-4 Mathematics Olympiad (2021)
- **Regional Contestant:** ICPC Asia Regional Contest **(2024, 2025)**

Education

Ho Chi Minh Open University **Graduated: Oct 2026**
Bachelor of Science in Computer Science ***(Major: Data Science)*** *Ho Chi Minh City*